[1] Let (X_{n_k}, Y_{n_k}) be an arbitrary sequence of (X_n, Y_n).

For (X_{n_k}), since $X_n \xrightarrow[n\to\infty]{P} X$, there exists a subsubsequence $X_{n_{k_j}} \xrightarrow[j\to+\infty]{a.s} X$. For $(Y_{n_{k_j}})$, there also exists a subsubsequence $(Y_{n_{k_{j_l}}})$ with $Y_{n_{k_{j_l}}} \xrightarrow[l\to\infty]{a.s} Y$. So $X_{n_{k_{j_l}}} + Y_{n_{k_{j_l}}} \xrightarrow[l\to\infty]{a.s} X+Y$

By Theorem 4.2.2, we have $X_n + Y_n \xrightarrow[n\to\infty]{P} X+Y$.

With same method, we can also prove that $X_n - Y_n \xrightarrow[n\to\infty]{P} X-Y$ and $X_n Y_n \xrightarrow[n\to\infty]{P} XY$.

[2] (a) By Theorem 4.3.1, we know $X_n \xrightarrow[n\to\infty]{P} Y$. So there exists subsequence $X_{n_k} \xrightarrow[k\to\infty]{a.s} Y$. Since $X_{n_k} \xrightarrow[k\to\infty]{a.s} X$, we obtain $X=Y$, a.e.

(b) Consider the probability space $(\Omega=[0,1), \mathcal{B}([0,1)), P=$ Lebesgue measure$)$.

Define $f_n(x) = \begin{cases} n & x\in(0,\frac{1}{n}] \\ 0 & x\in(\frac{1}{n},1) \end{cases}$ so $f_n(x) \xrightarrow[n\to\infty]{a.e} 0$.

But $E[f_n] = 1 \not\to 0$, as $n\to+\infty$. So $f_n \not\xrightarrow{L^P} 0$.

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(c) We also consider the same probability space as (b).

Define $f_1^{(1)}(x) = 1$,

$$f_1^{(2)}(x) = \begin{cases} 1, & x \in [0, \frac{1}{2}) \\ 0 & x \in [\frac{1}{2}, 1) \end{cases}, \quad f_2^{(2)}(x) = \begin{cases} 0 & x \in (0, \frac{1}{2}] \\ 1 & x \in (\frac{1}{2}, 1) \end{cases},$$

and

$$f_i^{(k)}(x) = \begin{cases} 1 & x \in (\frac{i-1}{k}, \frac{i}{k}) \\ 0 & x \notin (\frac{i-1}{k}, \frac{i}{k}) \end{cases} \quad (i = 1, 2, \cdots k)$$

Set $\varphi_1(x) = f_1^{(1)}(x)$, $\varphi_2(x) = f_1^{(2)}(x)$, $\varphi_3(x) = f_2^{(2)}(x)$, $\varphi_4(x) = f_1^{(3)}(x)$

$\varphi_5(x) = f_2^{(3)}(x)$, ...

In fact, $\varphi_n(x) \xrightarrow[n\to\infty]{a.s} 0$, because for each $x \in [0,1)$ and all $N \in \mathbb{N}_+$ there always exists some $N_0 > N$ such that $\varphi_{N_0}(x) = 1$.

However, $\int_0^1 f_i^{(k)}(x)\,dx \le \frac{1}{k}$, so $\varphi_n \xrightarrow[n\to\infty]{L^1} 0$.

3. (a) Let (X_{n_k}) be an arbitrary subsequence of (X_n).

Since $X_{n_k} \xrightarrow[k\to\infty]{P} X$, there exists a subsubsequence $(X_{n_{k_j}})$ s.t $X_{n_{k_j}} \xrightarrow[j\to\infty]{a.s} X$. But $|X_{n_{k_0}}| \le X \in L^1$, by DCT,

$$E[X] = E[\lim_{j\to\infty} X_{n_{k_j}}] = \lim_{j\to\infty} E[X_{n_{k_j}}].$$

So for any subsequence, there exists a subsubsequence with

$$E[X_{n_{k_j}}] \to E[X].$$

We must have $E[X_n] \to E[X]$. By Theorem 4.3.1, $X_n \xrightarrow{L^1} X$

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(b) Since $X_n \downarrow X$ and $X_n \in \mathcal{L}^1$, we have $0 \le X_1 - X_n \uparrow X_1 - X$.

By monotone convergence theorem, $E[X_1 - X_n] \uparrow E[X_1 - X]$.

So we have $E[X_n] \downarrow E[X]$. (*)

However $\inf E[X_n] > -\infty$, so $-\infty < E[X] \le E[X_1] < \infty$.

Then $X \in \mathcal{L}^1$. By (*) and $X_n \to X$ a.s., we have $X_n \xrightarrow{\mathcal{L}^1} X$.

4 Notice that

$$E[|X_n|^\alpha \cdot 1_{\{|X_n|^\alpha > M\}}] = E\left[\frac{|X_n|^\beta}{|X_n|^{\beta-\alpha}} 1_{\{|X_n|^\alpha > M\}}\right]$$

$$\le \frac{1}{M^{(\beta-\alpha)/\alpha}} E[|X_n|^\beta \cdot 1_{\{|X_n|^\alpha > M\}}]$$

$$\le \frac{1}{M^{(\beta-\alpha)/\alpha}} \sup_{n\ge 1} E[|X_n|^\beta].$$

we have $\sup_{n\ge 1} E[|X_n|^\alpha 1_{\{|X_n|^\alpha > M\}}] \le \frac{1}{M^{(\beta-\alpha)/\alpha}} \sup_{n\ge 1} E[|X_n|^\beta]$.

Thus

$$\lim_{M\to\infty} \sup_{n\ge 1} E[|X_n|^\alpha \cdot 1_{\{|X_n|^\alpha > M\}}] = 0.$$

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